Macroeconomic Effects of Foreign Direct Investment: Should We Worry about Offshoring?

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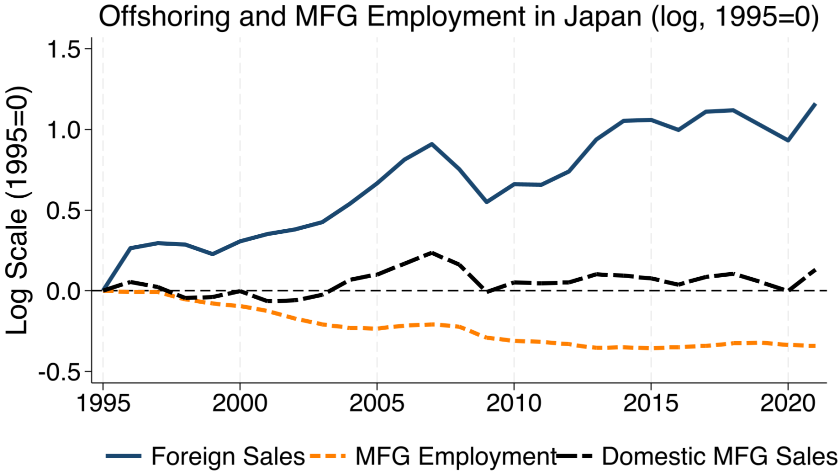
Preliminary; working paper expected in 5 weeks

Rise of Offshoring, Decline of Manufacturing Employment



Source: BSOBA, BSJBSA, JIP


47
MADE IN THE USA!

More Offshoring, Fewer Jobs? Should We Worry?

- Important and Controversial Topic
  - "One of the **most controversial** aspects of globalisation is offshoring."
    - Kiyota et al (2022, VOXEU)

- "**little agreement** among academic economists regarding the sign..."
  - Kovak et al (2021, RESTAT)

- Some evidence (in the US or Japan)
  - Parents' overall emp: $\uparrow$: Kovak et al (2021, RESTAT),...
  - Inequality: $\uparrow$: Feenstra and Hanson (1999, QJE), Hummels et al (2014, AER)
  - Regional/Sectoral emp: zero or $\uparrow$? Kovak et al (2021), Kiyota et al (2025)

This Paper

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- Employment
  1. Micro: Across Firms – Employment increases.
  2. **Macro: MFG Aggregate (Japan) – Employment decreases**
     - Confirm sizable substitution against domestic suppliers
     - Simple PE accounting: Net job loss of about 399K (rel. 1.26M affiliate jobs)
     - GE: exposed-sector loss remains negative (-688K), offset by nontradables
- GE: labor reallocates to nontradables; aggregate TFP falls; welfare rises (???)

Firm-level Evidence

Offshoring and Employment

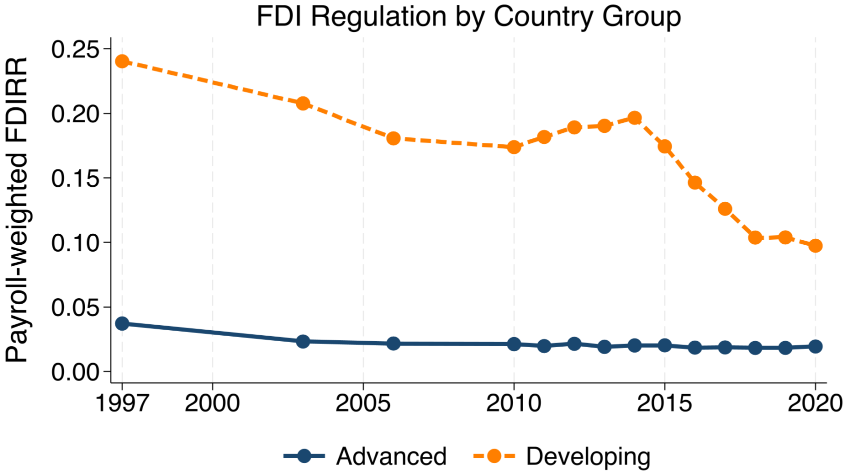
- Firm-level changes in employment on changes in offshoring

$$\Delta \ln L_i = \beta\, \Delta \mathrm{OFS}_i + \varepsilon_i$$

  - $\Delta\mathrm{OFS}_i$: Changes in firm *i*'s foreign affiliates' employment (BSOBA)

- Of course, this is problematic—Omitted Variable Bias
  - Positive demand/supply shock to the firm can increase both L and OFS

FDI Regulation Index by Country



Source: BSOBA, OECD

Foreign Affiliates by Host Country, 2019

Host country	Affiliates	%	Employment	%
China	5,240	26.4	1,254,477	22.3
United States	2,193	11.0	749,447	13.3
Thailand	1,960	9.9	641,793	11.4
Indonesia	965	4.9	402,053	7.1
Vietnam	956	4.8	389,887	6.9
Total	**19,885**	**100.0**	**5,626,167**	**100.0**

Source: BSOBA

Firm-level Analysis: Specification

- Long-difference between 2010-2019; with Shift-Share IV (SSIV); i firms

$$\Delta \ln L_i = \beta \, \Delta \text{OFS}_i + \varepsilon_i$$

  - ΔOFS_i: Changes in foreign affiliates' emp (BSOBA), rel. domestic emp (TSR)
  - SSIV

$$\Delta Z_i \equiv \sum_{c,s} \underbrace{\frac{Y_{i,c,s,t0}}{Y_{i,t0}}}_{\text{Share}} \cdot \underbrace{\Delta Z_{c,s}}_{\text{Shift}}$$

  - $\Delta Z_{c,s}$: Country-sector specific FDI Regulation Index (OECD)
  - $Y_{i,c,s,t}$: Payroll of firm i' affiliate in country c, sector s, time t (BSOBA)
  - Exposure-robust SEs

1st Stage: Looser Regulations Increase FDI

	Foreign Aff Emp Change/Baseline Domestic Emp			
	(1)	(2)	(3)	(4)
FDIRR exposure	0.78	-6.38***	-7.45***	-7.54***
	(0.77)	(1.40)	(1.52)	(1.53)
Host GDP trend		-3.72***	-3.34***	-3.30***
		(0.73)	(0.80)	(1.00)
Host RER trend			-1.14	-0.98
			(0.89)	(1.21)
China exposure				-0.09
				(0.69)
Obs.	2,669	2,669	2,669	2,669
First-stage F	1.05	20.93	24.17	24.27

Source: BSOBA, TSR, OECD

Reduced Form: Looser Regulations Increase Employment

	TSR parent domestic employment			
	(1)	(2)	(3)	(4)
FDIRR exposure	-0.11	-0.59***	-0.70***	-0.75***
	(0.11)	(0.16)	(0.21)	(0.21)
Host GDP trend		✓	✓	✓
Host RER trend			✓	✓
China exposure				✓
Obs.	2,669	2,669	2,669	2,669

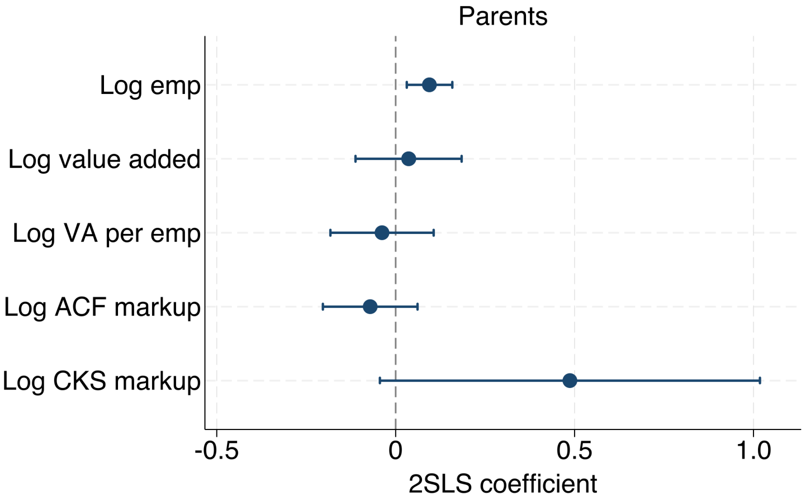
Source: BSOBA, TSR, OECD

2SLS Results: FDI Increases Parents' Employment

	TSR parent domestic employment			
	(1)	(2)	(3)	(4)
Affiliate employment change	-0.14	0.09***	0.09***	0.10***
	(0.20)	(0.03)	(0.03)	(0.03)
Host GDP trend		✓	✓	✓
Host RER trend			✓	✓
China exposure				✓
Obs.	2,669	2,669	2,669	2,669
First-stage F	1.05	20.93	24.17	24.27

Source: BSOBA, TSR, OECD

2SLS Results: No Further Effect



Source: BSOBA, TSR, OECD

Summary of Firm-level Empirical Results

- Revisit/replicate Kovak et al (2021) in Japan
- Confirm the key result (and the results are robust) across Firms

$$\text{corr}(\Delta\text{Offshoring}, \Delta\text{EMP}) > 0$$

Effects on Suppliers

Supplier-level Analysis: Specification

- Long-difference between 2010-2019

- Again, SSIV (long-difference); s supplier firms

$$\Delta \ln L_s = \beta \left[\sum_{i \in B(s)} \omega_{i,s} \cdot \Delta \mathrm{OFS}_i \right] + \varepsilon_{s,t}$$

- SSIV: Suppliers' exposure to buyer-weighted FDI regulation

$$\Delta Z_s = \sum_{i \in B(s)} \omega_{i,s} \cdot \Delta Z_i, \quad \Delta Z_i \equiv \sum_{c,s} \frac{Y_{i,c,s,t0}}{Y_{i,t0}} \cdot \Delta Z_{c,s}$$

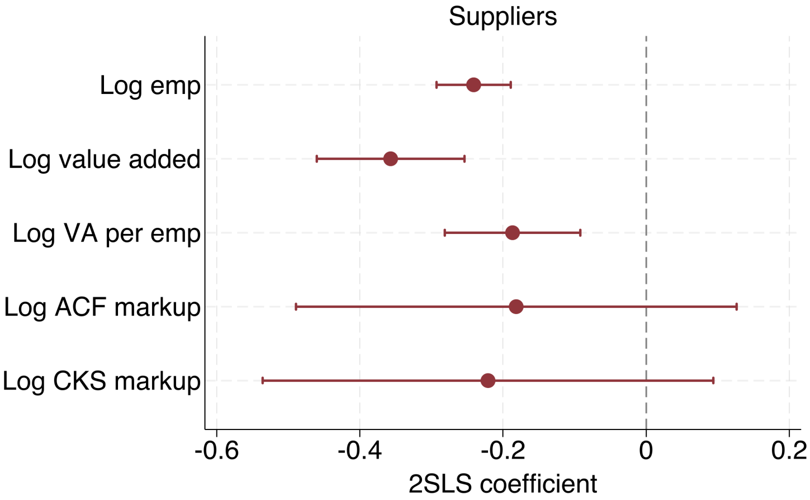
where $\omega_{i,s}$ is employment weight and ΔZ_i is our previous IV

2SLS: Buyers' FDI Decreases Suppliers' Employment

	Changes in supplier employment			
	(1)	(2)	(3)	(4)
Buyer FDI change	-0.18***	-0.24***	-0.24***	-0.24***
	(0.03)	(0.03)	(0.03)	(0.03)
Host GDP IV		✓	✓	✓
Host RER IV			✓	✓
Own FDIRR control				✓
Obs.	491,176	491,176	491,176	491,176
First-stage F	222.88	111.77	85.43	83.75

Source: BSOBA, TSR, OECD

2SLS Results: Suppliers' Size and Productivity Decline



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Simple Accounting: 2010-2019

- Before doing a full GE, try a simple PE accounting
    - ”How many jobs increase/decrease per predicted offshoring, $\widehat{\Delta OFS_i}$?”

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- Direct effect (firm/parent-level regression coef $\widehat{\beta^P}$)

$$\widehat{\Delta L^{\text{Direct}}} = \sum_i L_{i,t0} \cdot \exp\left(\widehat{\beta^P} \cdot \widehat{\Delta OFS_i}\right) \approx 121K$$

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- Net Effect: $\widehat{\Delta L^{\text{Direct}}} + \widehat{\Delta L^{\text{Indirect}}} \approx -399K$

Quantitative Model

Model Overview

Small open economy with two sectors (tradable vs non-tradable)

1. Tradable goods producers (=buyers) use domestic and foreign intermediates
    - Offshoring = substitution between domestic and foreign inputs
    - Heterogeneous in productivity, markup, and labor wedge

2. Non-tradable goods producers + Usual across-sector demand

3. Market clearing

Model 1/4. Tradable final-good producers

- Final assembly firm

$$Y^T = \left[M \int y(z)^{\frac{\varepsilon-1}{\varepsilon}} \, \mathrm{d}G(z) \right]^{\frac{\varepsilon}{\varepsilon-1}}, \qquad \varepsilon > 1,$$

- Variety Production (Firm i with z_i)

$$y_i = z_i \, \ell_i^{\alpha} x_i^{1-\alpha},$$

  where x_i is an intermediate (next slide)

- Heterogeneous Markup (μ_i)

$$p_i = \mu_i \, mc_i.$$

  where marginal costs embed labor wedges τ_i^L too

$$mc_i = \frac{C_\alpha}{z_i} \left[w(1+\tau_i^L) \right]^{\alpha} (P_i^x)^{1-\alpha}, \qquad C_\alpha \equiv \alpha^{-\alpha}(1-\alpha)^{-(1-\alpha)}.$$

Model 2/4: Intermediate Input Choices: Offshoring

- Non-FDI firms source only domestically: $x_i = x_i^D$
- FDI firms pay fixed costs f^F and can source from abroad:

$$x_i = \left[\omega^{1/\sigma} \left(x_i^D \right)^{\frac{\sigma-1}{\sigma}} + (1-\omega)^{1/\sigma} \left(x_i^F \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \qquad 0 < \omega < 1, \quad \sigma > 1,$$

- Foreign inputs available at the exogenous price

$$p^F = \tau \cdot \overline{p^F},$$

where τ is an offshoring wedge (our policy wedge)

Model 3/4. Non-Tradable and Household Demand

- Non-tradable Production

$$Y^N = z^N \ell^N,$$

- Households' demand for sectoral goods

$$U(C^T, C^N) = (C^T)^{1-\eta}(C^N)^{\eta},$$

s.t.

$$I = w\bar{L} + \Pi + \mathrm{Reb} + T, \qquad C^T = (1-\eta)I, \qquad p^N C^N = \eta I.$$

Model 4/4. Market Clearing and Equilibrium

- Domestic intermediate market $X^D = M \int x^D(z)\,\mathrm{d}G$,
- Nontradable market $Y^N = C^N$.
- Domestic labor market

$$\underbrace{M \int \ell(z)\,\mathrm{d}G}_{\equiv L^P} + \underbrace{f^F M(z^*)^{-\theta}}_{\equiv L^{HQ}} + \underbrace{\ell^S}_{\equiv L^S} + \underbrace{\ell^N}_{\equiv L^N} = \bar{L},$$

- An equilibrium consists of a wage, prices, and allocations such that
    - Producers solve profit max. (labor demand, offshoring, FDI entry)
    - Households maximize utility
    - Markets clear

Effects of Offshoring

Equilibrium Characterization 1/2: Input Substitution

Foreign input price falls with τ

$$\tau \downarrow \implies p^F \downarrow$$

For offshoring firms, total intermediate price falls

$$d \log P^x = s_F \, d \log p^F < 0$$

Their domestic input share falls

$$d \log s_D = (\sigma - 1) s_F \, d \log p^F < 0$$

Therefore, when τ decreases,

$$\boxed{P^x \downarrow, \qquad s_D \downarrow, \qquad s_F = 1 - s_D \uparrow}$$

Equilibrium Characterization 2/2: Firm-level responses

Let $\kappa \equiv (1-\alpha)(\varepsilon - 1)$ and $\chi_i \equiv -\Delta \log p_i^F > 0$. For an incumbent offshoring firm,

$$\Delta \log L_i^P = \kappa s_F \chi_i$$

so that

$$p^F \downarrow \implies \quad Y_i \uparrow, \; L_i^P \uparrow .$$

For domestic supplier input demand,

$$\Delta \log x_i^D = [\kappa - (\sigma - 1)] \, s_F \chi_i .$$

Hence domestic supplier demand falls iff

$$x_i^D \downarrow \iff \quad \sigma - 1 > \kappa .$$

Calibration: Two IV Moments, Two Elasticities

Block	Objects	Values	Discipline
Baseline	(s_F^0, q, η)	$(0.113,\ 0.241,\ 0.171)$	JIP/TSR; baseline clearing
IV targets	$(\theta_P^{IV}, \theta_S^{IV})$	$(0.094,\ -0.236)$	Parent and supplier 2SLS
Elasticities	(κ, σ)	$(0.830,\ 3.913)$	Analytic IV inversion
Production	$(\alpha, \varepsilon, \theta)$	$(0.919,\ 11.236,\ 5.863)$	Cost shares/tail moments
Wedge loadings	(ϕ_μ, ϕ_τ)	$(0.358,\ 0.172)$	Markup/MRPL-size slopes
Counterfactual	$\widehat{\tau}_1$	0.842	Match $+1.26$ M affiliates

Quantitative Results

Exercise.

- Choose τ_1 to reproduce FDIRR-predicted increase of 1.26 M affiliate jobs
    - $\tau : 1 \to 0.842$
- then solve the general equilibrium.

Results:

- Employment (reallocation)
- Productivity
- Welfare

GE Reallocates Labor to Nontradables

Employment outcome	PE Simple accounting	GE counterfactual
Parent firms	$+121$K	
Suppliers	-520K	
Exposed sector	**-399K**	
Nontradables	—	
Total domestic	—	

GE Reallocates Labor to Nontradables

Employment outcome	PE Simple accounting	GE counterfactual
Parent firms	$+121$K	$+75$K
Suppliers	-520K	-763K
Exposed sector	$-$**399**K	$-$**688**K
Nontradables	–	$+688$K
Total domestic	–	0, fixed $\bar{L}$

GE Reallocates Labor to Nontradables

Employment outcome	PE Simple accounting	GE counterfactual
Parent firms	$+121$K	$+75$K
Suppliers	-520K	-763K
Exposed sector	$-$**399**K	$-$**688**K
Nontradables	–	$+688$K
Total domestic	–	0, fixed $\bar{L}$

- The exposed-sector net loss remains negative in GE
- Nontradables absorb the displaced exposed-sector labor

Aggregate TFP: Technology Gain, Reallocation Drag

- Tradable Productivity: +0.21 pp

$$\underbrace{\Delta \log A_T}_{+0.21pp} = \underbrace{\Delta \log A_T^*}_{\text{technology/selection } (+0.20pp)} + \underbrace{\Delta \log AE}_{\text{allocative efficiency}(+0.01pp)}$$

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- Aggregate Productivity

$$A = \frac{A_M L_E + A_{NT} L_{NT}}{\bar{L}}, \qquad \frac{A_{NT}}{A_M} = 0.635.$$

$$\Delta L_E = -688\text{K}, \qquad \Delta L_{NT} = +688\text{K}.$$

$$\Delta \log A \approx \underbrace{+0.076}_{A_T \text{ gain}} \quad \underbrace{-1.092}_{\text{reallocation to } A_{NT}} \quad \underbrace{-0.005}_{\text{cross term}} \approx -1.021 \text{ pp}$$

Welfare Decomposition

$$
\begin{aligned}
d \log W \cdot I = \; & \underbrace{-X^F \, dp^F}_{\text{direct price fall}} \\
& \underbrace{+ \, M \int (\mu_i - 1) \left[w \, d\ell_i + p^D \, dx_i^D + p^F \, dx_i^F \right] dG_i}_{\text{markup reallocation}} \\
& + \underbrace{M \int \mu_i \tau_i^L w \, d\ell_i \, dG_i}_{\text{MRPL/labor-wedge reallocation}} + \underbrace{S(z^*) \, dn_F}_{\text{FDI selection}} .
\end{aligned}
$$

Welfare (work-in-progress)

Component	pp
Direct import-price	$+0.121$
Markup reallocation	$+0.124$
Labor-wedge/MRPL reallocation	$+2.842$
FDI selection	$+0.024$
Total	$+3.111$

- Labor-wedge/MRPL reallocation: employment increases in high MRPL firms
- Currently missing: distortions in non-tradables, limited sectoral mobility...
  - so the welfare numbers are work-in-progress

Conclusion

- Offshoring **decreases** (exposed sector) employment at the **aggregate level**
    - opposite to the **cross-firm** sign

$$\text{corr}(\Delta\text{Offshoring}, \Delta\text{EMP}) > 0$$

- Caution for extrapolating micro evidence to macro questions
- Measured aggregate (domestic) TFP decreases
- Welfare improves (work-in-progress)

Appendix

2SLS: Related Suppliers Survive

	Changes in supplier employment			
	(1)	(2)	(3)	(4)
Parent-child buyer FDI	0.63	0.54***	0.27***	0.27***
	(0.53)	(0.20)	(0.10)	(0.10)
Other buyer FDI	-0.20***	-0.26***	-0.27***	-0.27***
	(0.03)	(0.03)	(0.03)	(0.03)
Split host GDP IVs		✓	✓	✓
Split host RER IVs			✓	✓
Own FDIRR control				✓
Obs.	491,176	491,176	491,176	491,176
First-stage F, parent-child	5.91	6.74	7.23	7.96
First-stage F, other buyers	127.73	65.01	48.02	47.31

Source: BSOBA, TSR, OECD